DSAI4901 · CAPSTONE PROJECT PROPOSAL

# Paper-Distillation PPT Agent

## From generating to verifying

A slide generation system built around content traceability and deck-readiness

**CHAN Yee Tung**
Supervisor: Mr. Chen Xianwei
October 2026

**35**
papers reviewed

**8**
functional gaps

**1**
research direction

# Literature review: from 367 papers to 35

**367** arXiv candidates

**69** after relevance screening

**35** read in depth (30 agents + 5 education)

**18** with author-released code

18/35 release author code (each link verified by HTTP); the rest are paper-only or anonymous submissions

### WHAT I ACTUALLY DID

- 12 arXiv queries across slide, PPT, poster and deck
- Relevance screening to remove unrelated work
- Full-text keyword scan of all 35 papers
- GitHub links extracted from each PDF, then verified by HTTP

# Four things they already do well

First establish what is solved — this decides what I should not rebuild

01

## Generation & layout

Decks already look convincing: stable layout, consistent styling, competitive in blind tests.

Zheng et al. (2025); Vats et al. (2026)

02

## Native editable output

Work has moved into the .pptx object layer, preserving styling and editability.

Jung et al. (2025); Zheng et al. (2026b)

03

## Evaluation methods

From holistic scores to task-based tests: answering questions on a deck, gating readiness.

Pang et al. (2025); Hu et al. (2026)

04

## Multimodal & narrative

Figures, tables and equations enter the layout; story structure is now modelled explicitly.

Liang et al. (2025); Ozden et al. (2026a)

**Conclusion: generation is commoditised. Building one more generator is not a contribution.**

# Decomposing a PPT agent into 8 functions

A precise gap needs a function map — the next four slides walk through it

① **Parsing** → ② **Content** → ③ **Narrative** → ④ **Layout** → ⑤ **Output** → ⑥ **Inspection**

⑦ **Speaker notes** — Cross-stage: turn content into speakable pacing

⑧ **Provenance** — Cross-stage: every sentence needs a source

Next four slides: two gaps per slide — symptom · evidence · consequence

# Gap 1 | Parsing and content selection

① PARSING

## Figures and captions split apart

SYMPTOMS

- figure and caption are detached, so image and text no longer correspond
- two-column papers are read in the wrong order; paragraphs get spliced
- tables are flattened into a wall of text, losing row and column structure

EVIDENCE · APA

Parsing is treated as preprocessing and figure–text alignment is not enforced (Pang et al., 2025; Fu et al., 2021).

CONSEQUENCE

A corrupted input is inherited by every later step, and the error is invisible in the final deck.

② CONTENT SELECTION

## Even allocation, no emphasis

SYMPTOMS

- one slide per section, regardless of how much it matters
- the key experimental result is dropped while background survives
- method details are pushed into the introduction slide

EVIDENCE · APA

Most methods concentrate on text selection and ignore the link between space and importance (Liang et al., 2025; Chen et al., 2026).

CONSEQUENCE

The deck is complete but has no emphasis; the audience keeps only the title and the speaker cannot name the key claim.

# Gap 2 | Narrative and layout

③ NARRATIVE

## A table of contents, not a talk

SYMPTOMS

- section-by-section summarising reproduces the paper's order
- ten slides of background, two of results — emphasis inverted
- no motivation → problem → insight → evidence arc

EVIDENCE · APA

Source text is summarised straight into slides (Ozden et al., 2026a); explicit narrative planning remains rare (Vats et al., 2026).

CONSEQUENCE

The speaker reads a table of contents aloud and the audience cannot build causal understanding.

④ LAYOUT

## Overflow, overlap, clipping

SYMPTOMS

- text runs past the slide edge; elements overlap each other
- fonts too small and contrast too low to read once projected
- figures cropped, whitespace unbalanced, every slide a wall of bullets

EVIDENCE · APA

Local failures include overflow, overlap, clipping and off-canvas placement (Tian et al., 2026); models understand content yet fail at layout (Huang et al., 2025).

CONSEQUENCE

This is the most visible failure: text clipped and figures covered during the talk, so the audience loses trust immediately.

# Gap 3 | Output and inspection

⑤ OUTPUT

## Looks usable, cannot be edited

SYMPTOMS

- output is an image or HTML, not editable inside PowerPoint
- even .pptx output is all text boxes — editing one word breaks the layout
- charts are rasterised, so the underlying data is gone

EVIDENCE · APA

Agents frequently rasterise editable content or break the layout (Zheng et al., 2026b); only an object-model route preserves styling (Jung et al., 2025).

CONSEQUENCE

Users get an artefact they cannot modify, so it is unusable and the tool is abandoned.

⑥ INSPECTION

## One-shot feedback, no attribution

SYMPTOMS

- the whole deck is rewritten and then reviewed once
- errors cannot be localised to an element — which shape overflows?
- repair can make things worse, with no convergence guarantee

EVIDENCE · APA

A monolithic one-version-one-feedback loop makes local errors hard to attribute (Tian et al., 2026); reflection must be grounded in rendered state (Zheng et al., 2026a).

CONSEQUENCE

Iteration becomes a gamble: each pass may degrade the deck and nobody can say why.

# Gap 4 | Speaker notes and provenance

⑦ SPEAKER NOTES

## Notes are the bullets again

SYMPTOMS

- speaker notes duplicate the text already on the slide
- no time budget — no idea how many slides fit three minutes
- no transitions, so slides do not connect to each other

EVIDENCE · APA

Delivery is under-optimised in current systems (Yang et al., 2026); only 5 of 35 papers mention speaker notes, none as a research focus.

CONSEQUENCE

The speaker reads the slide while the audience reads and listens at once, splitting attention.

⑧ PROVENANCE

## Writes the words, cannot cite them

SYMPTOMS

- bullets carry no source, so there is no path back to a passage
- related work is presented as 'we propose'
- figures and tables misnumbered; numbers attached to the wrong claim

EVIDENCE · APA

Retrieval improves reliability but per-sentence provenance is not exposed (Xie et al., 2025); global intent is kept while sentence-level anchors are missing (Ozden et al., 2026a).

CONSEQUENCE

A wrong number goes unnoticed; in academic settings one misattributed claim destroys the credibility of the talk.

# A single failure is amplified downstream




Every upstream error is amplified downstream — and cannot be traced back

### Audience

Cannot follow or remember; sees clipped text live

→ Attention lost

### Speaker

Will not trust it; must rebuild; cannot answer 'where does this number come from?'

→ Trust collapses

### Field

Cannot compare systems, because there is no shared measure of trustworthiness

→ Progress stalls

**The shared problem across all 35 papers: everyone optimises generation; no one measures trust.**

# My direction: a paper-distillation PPT agent

Explicitly not lecture notes — the two tasks have different success criteria

| | Paper distillation (this project) | Lecture notes (not doing) |
|---|---|---|
| **Input** | A single paper (PDF) | Course material or lecture content |
| **Goal** | Compress the research into a defensible argument chain | Break knowledge into learning units |
| **Success** | Traceable content and a complete argument | Learner comprehension and exercises |
| **Hard part** | Factual correctness | Instructional design |

**RESEARCH QUESTIONS**

**RQ1** How can every bullet be anchored to a verifiable source?

**RQ2** How can deck usability be turned into a computable metric?

**RQ3** Can an inspect–repair loop improve both at an acceptable cost?

# Three contributions

## 01 A computable notion of trustworthiness

Content trust and deck usability each become an automatically measurable ratio: traceable-bullet ratio and deck-ready rate.

**Deliverable:**
**metric definitions + open-source toolkit**

## 02 A step-wise inspect–repair loop

Change one element at a time, render and verify immediately, so errors are attributable; report convergence rather than only a final score.

**Deliverable:**
**agent implementation + convergence curves**

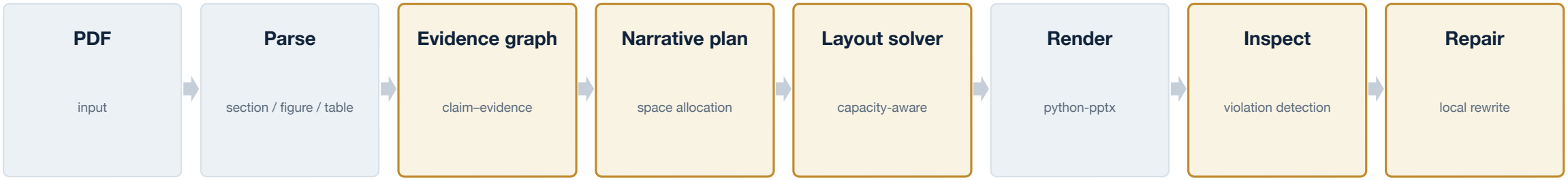
## 03 A public dataset and evaluation protocol

150 human-annotated reference slides, a 30-participant pairwise study, and rerunnable evaluation scripts.

**Deliverable:**
**dataset + benchmark script**

# System architecture

Gold = new contributions; the rest reuses validated practice




### Restrict the output space

The model writes structured content only; geometry and styling belong to templates

### One element at a time

Local rewrites with immediate verification make errors attributable

### Every judgement has a number

No 'looks fine' — only measured violations and ratios

The output is not an image but an editable .pptx: layout comes from constraint solving, content from the evidence graph.

# How I will show it works

Three tracks in parallel, avoiding the circularity of an LLM judging an LLM

| Track | Metric | Sample | Why it is needed |
|---|---|---|---|
| **Automatic (geometry)** | Violation count, deck-ready rate | 8 held-out papers, 150 annotated slides | Deterministic and rerunnable |
| **Task-based (QA)** | Answer accuracy from the deck | 10 questions × 8 papers | Tests whether a reader can extract correct facts |
| **Human** | Pairwise preference, speakability | 30 participants | Design has no ground truth, so only relative comparison is valid |

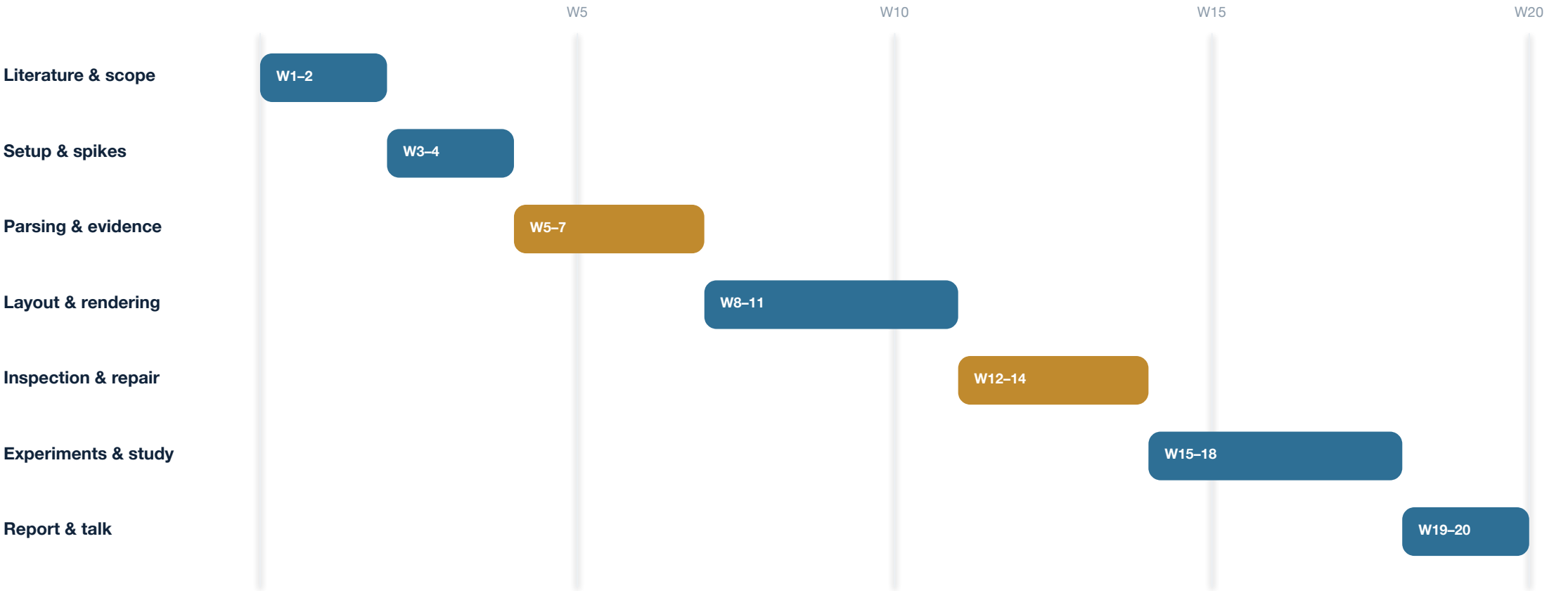
**BASELINES**

① Naive single-pass (one generation, no inspection)
② PPTAgent (strongest reproducible system)

**ABLATION**

Remove provenance anchors / remove inspection / remove repair
→ attribute each module's contribution

# 20-week timeline




**2026-11-05**

Proposal submission

**2027-01-07**

Interim Report

**2027-04-08**

Final Report + Turnitin + AI Report

# References


Chen, H., Zhou, X., Zhou, W., Shao, X., Zhu, Y., Wang, B., Hong, J., Jia, A., & Wu, F. (2026). X+Slides: Benchmarking audience-conditioned slide generation (arXiv:2606.19256). arXiv. https://doi.org/10.48550/arXiv.2606.19256

Fu, T.-J., Wang, W. Y., McDuff, D., & Song, Y. (2021). DOC2PPT: Automatic presentation slides generation from scientific documents (arXiv:2101.11796). arXiv. https://doi.org/10.48550/arXiv.2101.11796

Ge, J., Wang, Z. Z., Zhou, X., Peng, Y.-H., Subramanian, S., Tan, Q., Sap, M., Suhr, A., Fried, D., Neubig, G., & Darrell, T. (2025). AutoPresent: Designing structured visuals from scratch (arXiv:2501.00912). arXiv. https://doi.org/10.48550/arXiv.2501.00912

Hu, H., Dang, C., Liu, Y., Kang, H., He, C., & Li, W. (2026). PosterMELD: Multi-agent paper-to-poster generation for controllable design diversity with editable print-ready outputs (arXiv:2608.02218). arXiv. https://doi.org/10.48550/arXiv.2608.02218

Huang, Z., Liu, X., Hu, T., Zhang, K., & Liu, Y. (2025). PPTBench: Towards holistic evaluation of large language models for PowerPoint layout and design understanding (arXiv:2512.02624). arXiv. https://doi.org/10.48550/arXiv.2512.02624

Jung, K., Cho, H., Yun, J., Yang, S., Jang, J., & Choo, J. (2025). Talk to your slides: High-efficiency slide editing via language-driven structured data manipulation (arXiv:2505.11604). arXiv. https://doi.org/10.48550/arXiv.2505.11604

Liang, X., Zhang, Z., Zhang, X., Su, H., Xu, Y., Sun, S., & You, C. (2025). SlideGen: Collaborative multimodal agents for scientific slide generation (arXiv:2512.04529). arXiv. https://doi.org/10.48550/arXiv.2512.04529

Ozden, T. C., VS, S., Horoz, F., Kara, O., Kim, J., & Rehg, J. M. (2026a). Narrative-driven paper-to-slide generation via ArcDeck (arXiv:2604.11969). arXiv. https://doi.org/10.48550/arXiv.2604.11969

Ozden, T. C., VS, S., Horoz, F., Kara, O., Hakkani-Tür, D., Kim, J., & Rehg, J. M. (2026b). ConvDeck: Conversational paper-to-slide generation via stage-specific user feedback (arXiv:2609.00226). arXiv. https://doi.org/10.48550/arXiv.2609.00226

Pang, W., Lin, K. Q., Jian, X., He, X., & Torr, P. (2025). Paper2Poster: Towards multimodal poster automation from scientific papers (arXiv:2505.21497). arXiv. https://doi.org/10.48550/arXiv.2505.21497

Sun, E., Hou, Y., Wang, D., Zhang, Y., & Wang, N. X. R. (2021). D2S: Document-to-slide generation via query-based text summarization (arXiv:2105.03664). arXiv. https://doi.org/10.48550/arXiv.2105.03664

Tian, M., Zeng, Z., Yang, Y., Gao, X., Li, Y., Huang, Z., Wang, X., Lv, C., Cheng, M., Liu, B., Qiu, K., Dai, Q., Chen, D., Dong, Y., Zheng, X., Li, J., & Luo, C. (2026). ReDeck: Step-level render-grounded refinement for document-to-slide generation (arXiv:2609.00194). arXiv. https://doi.org/10.48550/arXiv.2609.00194

Vats, V., Sharma, K., & Wang, Y. (2026). SlideLab: Audience-centered scientific slide generation and evaluation (arXiv:2609.30294). arXiv. https://doi.org/10.48550/arXiv.2609.30294

Xie, E., Waterfield, D., Kennedy, M., & Zhang, A. (2025). SlideBot: A multi-agent framework for generating informative, reliable, multi-modal presentations (arXiv:2511.09804). arXiv. https://doi.org/10.48550/arXiv.2511.09804

Yang, M., Zhang, Z., Li, J., Liu, H., Cai, Y., & Zheng, W. (2026). DeepSlide: From artifacts to presentation delivery (arXiv:2605.15202). arXiv. https://doi.org/10.48550/arXiv.2605.15202

Zheng, H., Guan, X., Kong, H., Zheng, J., Zhou, W., Lin, H., Lu, Y., He, B., Han, X., & Sun, L. (2025). PPTAgent: Generating and evaluating presentations beyond text-to-slides (arXiv:2501.03936). arXiv. https://doi.org/10.48550/arXiv.2501.03936

Zheng, H., Mo, G., Yan, X., Yuan, Q., Zhang, W., Chen, X., Lu, Y., Lin, H., Han, X., & Sun, L. (2026a). DeepPresenter: Environment-grounded reflection for agentic presentation generation (arXiv:2602.22839). arXiv. https://doi.org/10.48550/arXiv.2602.22839

Zheng, H., Wang, F., Xiong, T., Yu, Y., Qian, S., Yu, H., Schwing, A., Nahrstedt, K., & Wu, M. (2026b). SLIDEFORGE: An LLM agent for controllable editing of slides as structured artifacts (arXiv:2609.03109). arXiv. https://doi.org/10.48550/arXiv.2609.03109